LIGHT LOOP V2.545.0 PACK | 00. Start Here: Builder & SW Guide

01. ElectroCookie Shield Explorer | 02. Horizontal Tube Pad Map | 03. 51-Step Solder Checklist | 04. 5m Master Vertical Blueprint | 05. Split Connectors (Opt A) | 06. Hybrid 2P+6D Connector (Opt B)

SUITE V2.545.0 | 8-TAP • 72.4 FPS • 3,600 LEDS (2 TUBES × 2 STRIPS) | RENESAS RA4M1 + ESP32-S3 DUAL-MCU

Light Loop 26 — Hardware Soldering & Software Architecture Hub

Welcome! Everything in this folder is organized so we can knock out the **ElectroCookie Proto Shield REV3.3** and **5-Meter Opposite-Strip Tube Soldering** while giving you the complete technical picture of how the C++ firmware, cycle-counted ARM assembly driver, and React Studio app fit together.

★ BENCH STATION A • SHIELD SOLDERING

1. ElectroCookie Proto Shield Interactive Explorer

Interactive 4-Tap vs. 8-Tap board explorer. Click any pin (D2–D12), transistor (Q1–Q6 PN2222A), resistor (100Ω, 330Ω, 22Ω), or 3-hole silver bar to isolate exact holes and step through soldering order.

OPEN INTERACTIVE SHIELD EXPLORER →

★ TUBES 1 & 2 • STRIP PAD MAP

2. Horizontal Compressed Tube Connection Map

Empty LED spans compressed out so only the 7 solder connection zones are shown left-to-right. Shows the 5-wire center channel, 3-Tap vs. 5-Tap power pads, and the +4 pad `wire_lock` GND offset.

OPEN HORIZONTAL PAD MAP →

★ 51 STEPS • LIVE BLE LED TESTER

3. Interactive 51-Step Soldering Checklist

Station-by-station soldering checklist with wire cut lengths, resistor specs, and live Web Bluetooth **LIGHT LED** buttons that illuminate

TRUE-SCALE ELEVATION

4. Master 5-Meter Vertical Straight-Line Blueprint

Full system schematic: Mean Well LRS-200-5 (40A) PSU, 6-way blade fuse block, Arduino

each solder joint on the strip to verify continuity immediately.

OPEN 51-STEP SOLDER CHECKLIST →

UNO R4 WiFi, wire cut-lengths, and both 5.0m (1,800-LED) tubes laid out side by side.

OPEN 5M MASTER BLUEPRINT →

HARNESS OPTION A

5. Split Connectors (2-Pin Power + 3-Pin Data)

Modular waterproof pigtail architecture separating high-current 14–16 AWG power feeds from 22 AWG data lines at the enclosure bulkhead.

OPEN OPTION A BLUEPRINT →

HARNESS OPTION B

6. Hybrid 2P+6D Single-Plug Connector

Consolidated 8-pin (2 large power pins + 6 signal pins) quick-disconnect bulkhead connector layout for single-click tube attachment.

OPEN OPTION B BLUEPRINT →

1. WHERE WE ARE & THE ELECTRICAL "WHY" BEHIND WHAT WE'RE SOLDERING

Physical Tube & Strip Topology (3,600 Total LEDs)

- **2 Tubes × 2 Opposite Strips**: Each 5.0m tube contains two 900-LED WS2812B strips mounted back-to-back (**Strip A Forward ►** `LED 0..899` and **Strip B Reverse ◄** `LED 1799..900`).
- **3-Turn Coiled Hoop Geometry (** `coilCount = 3` **)**: Each 5m tube coils into 3 overlapping 300-LED turns. Firmware projects animations across all 3 turns in lockstep via `applyPixelCoiled()`.
- **Why We Cut** `DIN` **at LED 450 (8-Tap Mode = 72.4 FPS)**: At 800 kHz (`30 µs/LED`), shifting 900 LEDs takes `27.0 ms` (~36.2 FPS). Cutting the copper `DIN` trace between LED 449 and 450 (while keeping `+5V` and `GND` continuous) and feeding LED 450 from a mid-tap pin halves the shift length to **450 LEDs (** `13.5 ms` **= 72.4 FPS)**.
- **Why** `GND` **Is Offset by +4 Pads from** `+5V`: Along the strip, every power injection tap solders `+5V` at Pad *N* and `GND` at Pad *N+4* (e.g., Pads `6/10`, `223/227`, `448/452`, `673/677`, `890/894`) so the wires enter the 3D-printed `wire_lock` channel in staggered slots without overlapping or shorting.

Why the ElectroCookie Shield Uses 6× PN2222A + 2× Direct 330Ω

- **Short Base Lines (** `D2 → T1-A` **,** `D4 → T2-A` **, <15 cm)**: Enter the base of the tube right next to the enclosure. Low wire capacitance allows a direct `330 Ω` **series damping resistor** on Column 10.
- **Long Mid-Tap & Far-End Lines (** `D3, D5, D6, D7, D11, D12` **, 2.5m–5.0m runs)**: Running 800 kHz square waves 2.5m to 5.0m up the internal tube channel requires high current drive and

source termination. Each channel uses a `PN2222A` **NPN Emitter-Follower**:

1. `100 Ω` from Arduino GPIO pin → Base 3-hole bar.
2. `PN2222A` Collector (Pin 3) → Center `+5V` bus; Base (Pin 2) → Base bar; Emitter (Pin 1) → Emitter bar.
3. `330 Ω` pull-down from Emitter bar → Center `GND` bus (ensures fast fall times for the 12-cycle `0.25 µs` T0H bit window).
4. `22 Ω` series output termination from Emitter bar → Output bar → Tube data wire.

Complete 8-Channel Pin & Tap Assignment Matrix

ARDUINO PIN	TAP BADGE	TUBE & STRIP SEGMENT	LED RANGE	MODE	ELECTROCOOKIE SHIELD STAGE
`D2`	`T1-A`	Tube 1 • Strip 1A Base (►)	`0 .. 449`	4-Tap & 8-Tap	Direct `330 Ω` (Col 10 Lower Row)
`D3`	`T1-B`	Tube 1 • Strip 1B Mid-Tap (✂ ►)	`450 .. 899`	4-Tap & 8-Tap	`Q6` PN2222A (`100Ω` + `330Ω` + `22Ω` , Lower Row)
`D5`	`T1-C`	Tube 1 • Strip 2A Far End (◄)	`900 .. 1349`	4-Tap & 8-Tap	`Q5` PN2222A (`100Ω` + `330Ω` + `22Ω` , Lower Row)
`D7`	`T1-D`	Tube 1 • Strip 2B Mid-Tap (◄ ✂)	`1350 .. 1799`	8-Tap Only	`Q3` PN2222A (`100Ω` + `330Ω` + `22Ω` , Upper Row)
`D4`	`T2-A`	Tube 2 • Strip 3A Base (►)	`1800 .. 2249`	8-Tap Only	Direct `330 Ω` (Col 10 Upper Row)
`D11`	`T2-B`	Tube 2 • Strip 3B Mid-Tap (✂ ►)	`2250 .. 2699`	8-Tap Only	`Q2` PN2222A (`100Ω` + `330Ω` + `22Ω` , Upper Row)
`D6`	`T2-C`	Tube 2 • Strip 4A Far End (◄)	`2700 .. 3149`	4-Tap & 8-Tap	`Q4` PN2222A (`100Ω` + `330Ω` + `22Ω` , Lower Row)
`D12`	`T2-D`	Tube 2 • Strip 4B Mid-Tap (◄ ✂)	`3150 .. 3599`	8-Tap Only	`Q1` PN2222A (`100Ω` + `330Ω` + `22Ω` , Upper Row)

2. PROGRAM LAYOUT & CODEBASE MAP (V2.545.0)

Embedded Microcontroller Firmware (`firmware/`)

- `firmware/light_loop_900/light_loop_900.ino` & `ws2812_parallel.S`:
  Main **Renesas RA4M1 (48 MHz Cortex-M4F)** firmware. Drives 40 procedural & physics modes, the Universal Landing Overlay Compositor (L1–L12), sub-microsecond `ARM_DWT_CYCCNT` parallel port-register WS2812 output, non-blocking 10–20 Hz sensory bus (VL53L5CX 64-zone ToF LiDAR + HC-SR04 Ultrasonic + DS1307 RTC), and `.noinit` SRAM crash telemetry.
- `firmware/esp32_unified_bridge/esp32_unified_bridge.ino`:
  Dual-Core FreeRTOS firmware for the onboard **ESP32-S3 radio co-processor**. **Core 0** runs an offline Wi-Fi Captive Portal (`SSID: LightLoop-Setup` , `192.168.4.1`) serving the 311.5 KB gzipped React PWA directly from flash ROM with zero internet required. **Core 1** runs the Coded PHY BLE GATT peripheral (`LightLoop-545`) bridged over 115,200-baud `Serial2` to the RA4M1.
- `firmware/bench_harness_tester/bench_harness_tester.ino`:
  Dedicated bench test sketch for Continuity / Channel / Tap verification during soldering.

React + TypeScript Web & Mobile Suite (Root)

- `index.tsx`: Root router (`?view=studio` , `?view=phone` , `?view=checklist` , `?view=classic`) and 4-phase Web Bluetooth post-connect settling state machine (`+550ms, +900ms, +1100ms, +1400ms`).
- `LightLoopAdaptiveStudio.tsx`: Desktop & tablet studio with concentric 3-coil 60 FPS canvas simulator, 40-mode matrix, 7 live modulators, and 5-zone scrubber.
- `MobilePhoneApp.tsx`: Mobile & Google Pixel Fold controller with single-row precision gesture scrubber and Android native BLE Coded PHY bridge support.
- `WiringChecklistApp.tsx`: Powers `/?view=checklist` and `checklist.html` —syncs soldering progress (64-bit mask) to Arduino Data Flash and fires `CHK` packets to light individual solder-joint LEDs.
- `LedController.ts`: Browser 60 FPS simulation engine mirroring all 40 C++ firmware modes and particle physics 1:1.
- `scripts/`: Hardware-in-the-loop CLI tools (`read-serial.py` , `scan-ble.py` , `test_ble.py` , `build-firmware.sh`).

> **Included Reference Files in This Google Drive Folder:** `00_START_HERE_BUILDER_GUIDE.md` • `light_loop_features.md` (Full Feature Catalog) • `unified_esp32_firmware_handoff.md` (ESP32-S3 Dual-Core Spec) • `four_line_segment_driver.md` (ARM DWT Assembly Timing Spec) • `power_protection_and_timing_precision.md` (Inrush Slew & Power Isolation) •

`electrocookie_photorealistic_wiring.png` · `full_board_photo_guide.png` · `electrocookie_light_loop_26.diylc`